Midel: Is Training the Deep Network Worth It?

The Honest Short Answer

Yes, continuing to train is worth it — but not because the deep network will replace the facts library. It's worth it because the two systems do fundamentally different things, and both are needed if your goal is a model that **behaves like a real AI**.

What Each System Actually Does Today

The `_facts` Library (Retrieval)

- Stores curated sentences: `{ id, question, text, category }`
- Answers by **keyword matching —** the `science_library.js` matcher scores lessons by topic-word overlap (≥72% coverage = answer)
- The chat shows **the exact stored sentence —** verbatim, never changed
- Can grow infinitely: just add rows to `taught_facts.js` or any `*_facts.js`
- **Zero training cost —** add a fact, reload the page, it works immediately

The Deep Network (Trained Weights)

- 4-layer transformer, 128-wide, ~1.68M weights in `stage3_brain.js`
- Learns to **continue** a prompt — given "what is a penguin", it generates the stored lesson tokens
- Used for: Signal/Attention panel, yes/no voting (`claimVote`), and **`lessonNetworkAnswer`** (network-written paraphrase when it's complete enough)
- Training cost: `--learn-new` takes minutes per lesson, with regression gates

What the Facts Library CANNOT Do

Here's the critical point — the facts library is a **lookup table**, not intelligence:

Capability	Facts Library	Trained Network
Answer a known question	✅ (keyword match)	✅ (continuation)
Answer a **rephrased** question	⚠️ (fuzzy match, but brittle)	✅ (learned representations)
Say **yes/no** to a claim ("Is Antarctica hot?")	❌ (no reasoning)	✅ (`claimVote` — learned from training)
Continue a thought / generate text	❌	✅ (greedy decode)
Generalize across topics	❌ (each fact is isolated)	⚠️ (limited, but real)
Show **attention patterns** (what the model "thinks")	❌	✅ (the whole right panel)
Handle a question **not in the library**	❌ (says "not known")	⚠️ (may produce related content)

The yes/no gate is the clearest example

When your model answers "Is the Sahara hot?" → **yes** and "Is Antarctica the largest **hot** desert?" → **no**, that reasoning is **only in the weights**. The facts library has the Sahara/Antarctica lesson, but it cannot parse a yes/no question, compare clauses, and decide. The deep network learned this distinction through the `claim_examples()` training, where it was shown:

```
<fact sentence> is the sahara hot ? yes
<fact sentence> is antarctica the largest hot desert ? no
```

That's genuine learning — the weights encode the **relationship** between subject and predicate, not just a stored string.

What the Deep Network CANNOT Do (Yet)

Be honest about the limits too:

1. **It cannot freely generate** like GPT/Gemini — it's ~1.7M params vs. billions. Its continuations are filtered to stay within the lesson's vocabulary (`filterProducedToLesson`).
2. **It forgets** — the replay window is 64 lessons. Older installed lessons can drift without being tested.

3. **It's word-level** — no subword tokenizer means unknown words are invisible.
4. **It's greedy** — no temperature/sampling, just argmax.

So Is Training Worth the Time and Energy?

✅ YES, for these reasons:

1. **Reasoning that lookup can't do.** Yes/no claims, attention patterns, and continuations are impossible with just `_facts.js` files. These are the behaviors that make Midel feel "intelligent" rather than being a search engine.
2. **The Signal/Attention panel.** This is Midel's showcase — the visual proof that a neural network is processing the question. Without installed weights, the right panel stays empty for that lesson. This is the **educational** and **demo** value of your project.
3. **Network-written answers.** When `lessonNetworkAnswerComplete` passes, the deep network can show its own paraphrase of the answer — demonstrating that it **understood** the content, not just stored it. This is the difference between a database and a model.
4. **Your stated north star.** From : *"Move step by step toward behaviors people associate with modern LLMs — broad knowledge, stable reasoning on taught facts, fluent continuations"* — you can't reach that with lookup alone.

⚠️ BUT keep these in mind:

1. **The library is still the primary answerer** — and that's fine. Even GPT-4 uses retrieval (RAG) before generation. Your `library-first` design is actually the **correct architecture** for factual Q&A. The ROADMAP explicitly says: *"Replacing the library with pure neural generation for factual Q&A"* is an **anti-goal**.
2. **Don't expect the 1.7M-param network to become a free-form conversationalist.** That would require orders of magnitude more parameters and pretraining data (Phase D in your roadmap). The deep network's job is to **support** the library, not replace it.
3. **The training loop ROI decreases over time.** The first 200 lessons teach the network the most. After 800+, each new lesson adds diminishing capability while increasing forgetting risk. This is normal for small models.

The Path Forward (What Would Make It Smarter)

Based on your ROADMAP phases, in order of impact:

Priority	What	Why it matters
1	**Keep feeding the facts library** (wiki batch, manual packs)	Immediate chat improvement, zero training cost
2	**Keep running `--learn-new` loop** at a comfortable pace	Signal/Attention panel, yes/no reasoning, demo value
3	**Phase B: Better retention**	More lessons stick without forgetting old ones
4	**Phase C: Subword tokenizer + bigger model**	Handles acronyms, rare words, richer continuations
5	**Phase D: Pretraining on text corpus**	This is where the model would start feeling "like a real LLM"

> IMPORTANT
>
> **The facts library and the deep network are not competing approaches — they're complementary layers.** The library gives Midel **breadth** (thousands of topics, instantly). The deep network gives Midel **depth** (reasoning, continuation, attention visualization). A real AI model needs both.

Bottom Line

Keep training. It's worth it. But don't think of it as "moving knowledge from facts files into weights." Think of it as:

- **Facts files** = what Midel **knows** (the encyclopedia)
- **Deep weights** = how Midel **thinks** (the neural patterns)

A smart model needs both. The training loop is what gives Midel its neural character — without it, you have a search engine with a nice UI. With it, you have a genuine micro-transformer that demonstrates real AI behavior: attention, continuation, yes/no reasoning, and learning.

The energy cost is modest (CPU minutes per lesson, not GPU hours), and each installed lesson makes the Signal/Attention panel richer. That's the visible proof that your model is more than a lookup table.